



OPENAI SOLUTIONS ENGINEERING

From fragmented supply-chain signals to governed decisions

A proposed OpenAI platform approach for Zeiss Supply Chain Management and IT leadership



TL;DR

OpenAI becomes the governed decision layer between ZEISS supply-chain data and operational action.

| Executive chat experience

| Enterprise integration and control layer

Problem

Context is split across SAP, logistics, suppliers and files.

Platform move

A central communication interface gives leaders **one decision surface**; the API integrates tools and controls.

Pilot

Start with one high-value workflow and validate faster triage, evidence quality and governed approvals.



Status quo and validation needs

Known operating context



- Advanced manufacturing, medical technology, quality and optics domains
- Decisions draw on SAP, supplier, logistics and document context
- Value depends on trustworthy answers, not another dashboard

Hypotheses from the technical discovery

- **Time to decision:** Teams lose time reconciling conflicting SAP, supplier, logistics and workbook updates before they can act.
- **Data confidence:** Leaders cannot immediately verify which source is current, permitted and backed by traceable evidence.
- **Governed action:** Mitigation slows when ownership and approval paths are unclear, increasing expedite spend, schedule churn and customer-delivery risk.



Providing a decision framework for existing data

SAP

- POs
- Material master
- MRP exceptions

Warehouse

- EWM stock
- Goods receipts
- Production buffers

Carriers

- DHL / FedEx
- ETA changes
- Customs events

Suppliers

- Commit dates
- Capacity
- Quality status

Files

- Scorecards
- Excel trackers
- Contracts

PAIN

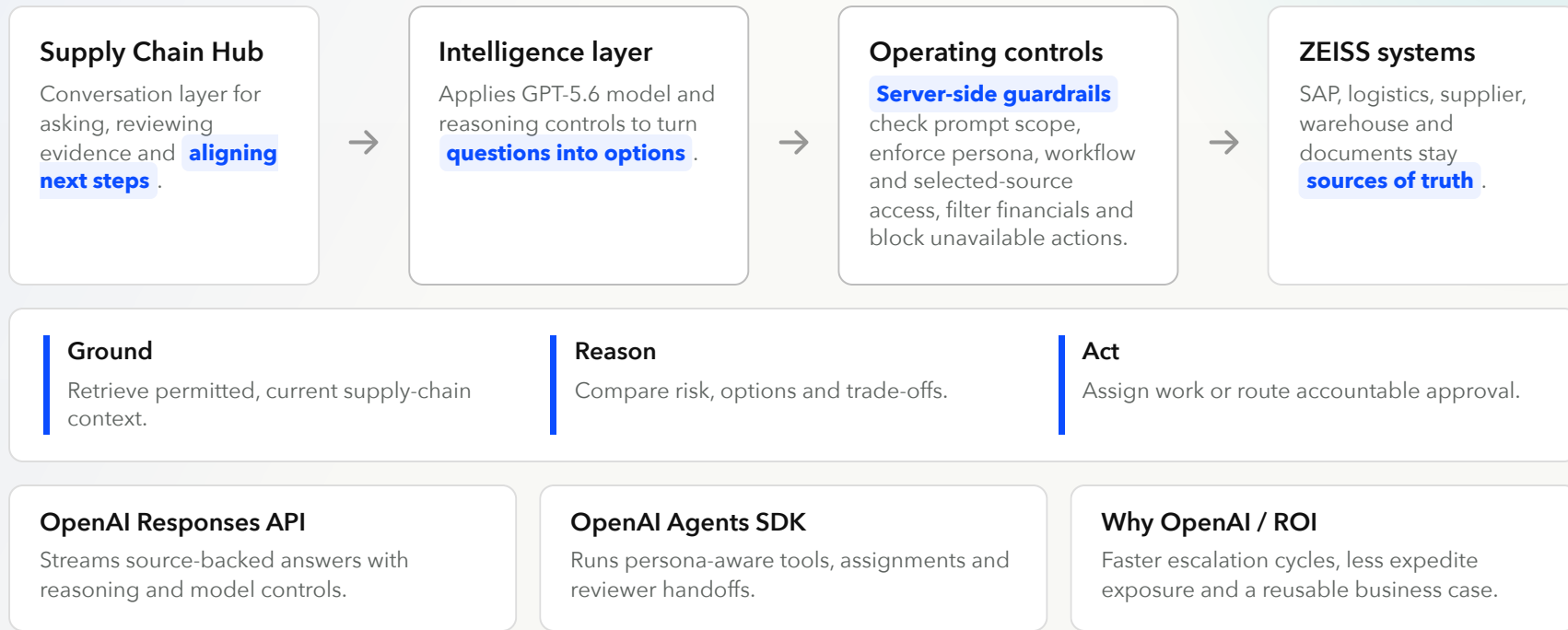
- Manual effort to reconstruct one trusted view
- Answer accuracy varies by source snapshot
- Low precision for lead-time and supplier risk
- Reliability depends on scarce experts

COST

- Expedite spend and buffers absorb uncertainty
- Leadership time shifts into escalation meetings
- Late mitigation creates schedule churn
- Scattered evidence increases quality exposure

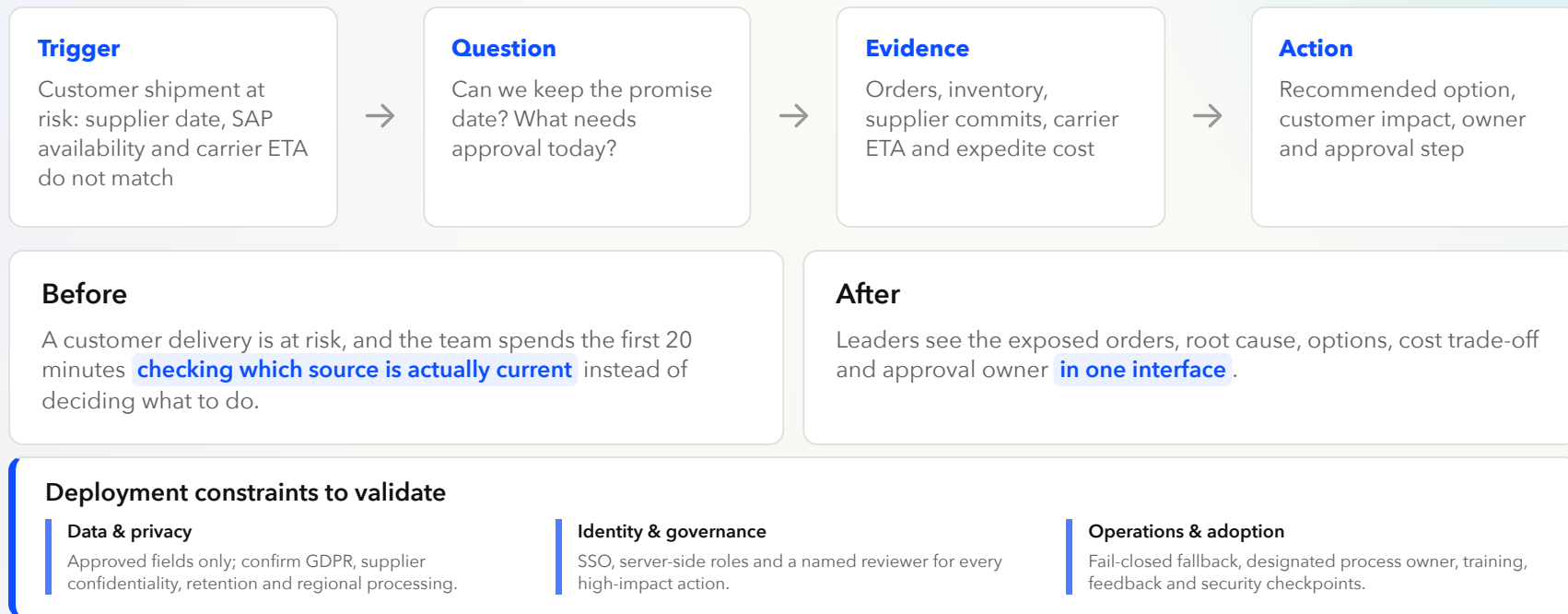


Solution overview





Executive workflow



Demo



Impact validation and success criteria

Validate value and control before scaling one workflow.

1

Annual value hypothesis · one workflow

€0.2M–€0.3M

Faster risk review

Time saved per completed case

Fewer urgent expedites

Avoided cases × actual expedite cost

Lower disruption exposure

Avoided incidents × verified impact

2

POC success criteria

Test routine cases, high-impact exceptions and permission boundaries.

Process value

≥25% faster review · ≥80% useful

How measured Compare timestamped case duration and collect a post-case user rating.

Decision quality

≥90% source-backed · <5% serious risks missed

How measured Review outputs against expert-approved scenarios and source evidence.

Technical reliability

≥95% correct approved system and workflow

How measured Inspect tool-call and trace logs across the scenario set.

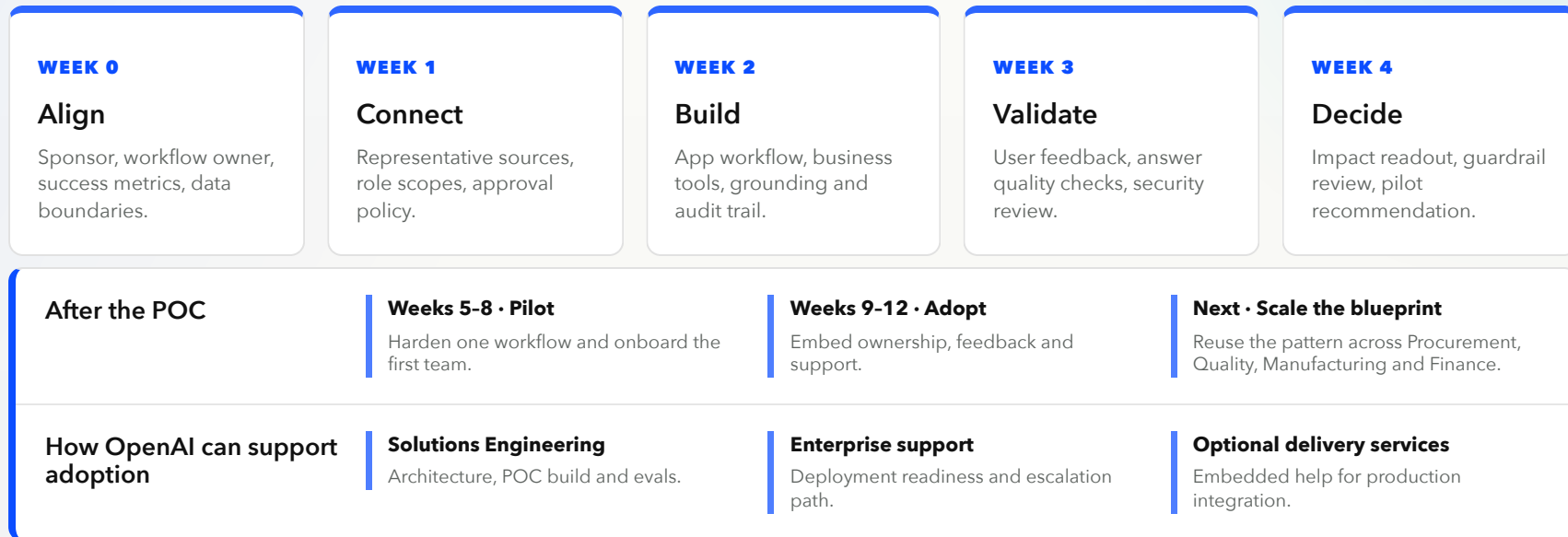
Governance

100% human review for high-impact actions

How measured Verify every required approval in the audit log.



Time to value



Annex



Actual Supply Chain Hub solution stack

Application experience

- Next.js App Router, React and strict TypeScript
- AI SDK React chat, persona-aware tools and source controls
- GPT-5.6 Sol, Terra and Luna with selectable reasoning effort

OpenAI Responses API

- Streams grounded answers through `openai.responses( ... )`
- Combines server-owned context, approved tools and reasoning controls
- Uses deterministic demo fallbacks when no live API key is configured

OpenAI Agents SDK

- Runs role-aware action tools and reviewer handoffs
- Enforces persona, workflow and source eligibility server-side
- Flushes traces after each action workflow for operational visibility

Guardrails, visuals and delivery

- `gpt-5.4-nano` classifies prompt scope with structured output and guardrails fail closed
- OpenAI image generation runs only when no trusted chart fits
- Vitest and strict typecheck; OpenNext and Wrangler deploy to Cloudflare Workers